CRITERION: LOCALLY OF FINITE TYPE AND INJECTIVE ON EVERY ETALE BASE CHANGE FORCES UNIVERSAL INJECTIVITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Universal injectivity is etale local on the target. Morphisms of Spaces, Lemma labelled spaces-morphisms-lemma-universally-injective-local; invoked at lines 225-227. Reduces the claim to the case where Y is a scheme.
- Conditions on a point of a fibre of a locally finite type morphism. Decent Spaces, Lemma labelled decent-spaces-lemma-conditions-on-point-in-fibre-and-qf; invoked at lines 229-231. Produces local quasi-finiteness of f from the injectivity assumption over a decent base.
- A locally quasi-finite morphism over a field has a scheme as source. Spaces over Fields, Lemma labelled spaces-over-fields-lemma-locally-quasi-finite-over-field; invoked at lines 233-235. Turns the fibre X_y into a scheme, so scheme-theoretic language becomes available.
- Every finite separable residue field extension is realised by an etale scheme over the base. More on Morphisms, Lemma labelled more-morphisms-lemma-realize-prescribed-residue-field-extension-etale; invoked at lines 242-243. Gives that the single-point property of X_y survives arbitrary finite separable extension of the residue field.
- Criterion for geometric connectedness by testing finite separable extensions. Varieties, Lemma labelled varieties-lemma-characterize-geometrically-disconnected; invoked at line 245. Converts the one-point fibre plus survival under finite separable extension into geometric connectedness.

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- Radicial morphisms are universally injective. Same chapter, Lemma labelled lemma-radicial-implies-universally-injective; invoked at line 255. Closes the argument from the purely inseparable residue field extension.

Source reference. Original lemma 0ACB: sources/stacks-chow/tags.txt, source lines 13049–13049. Navigation pointer for \ref{decent-spaces-lemma-conditions-on-point-in-fibre-and-qf} used inside the counted span (line 230); the argument that consumes it is retained in the span.

Source reference. Original lemma 0482: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 68–68. Navigation pointer for \ref{lemma-radicial-implies-universally-injective} used inside the counted span (line 255); the argument that consumes it is retained in the span.

Source reference. Original lemma 02LF: sources/stacks-chow/tags.txt, source lines 3406–3406. Navigation pointer for \ref{more-morphisms-lemma-realize-prescribed-residue-field-extension-etale} used inside the counted span (line 243); the argument that consumes it is retained in the span.

Source reference. Original lemma 03MX: sources/stacks-chow/tags.txt, source lines 4765–4765. Navigation pointer for \ref{spaces-morphisms-lemma-universally-injective-local} used inside the counted span (line 227); the argument that consumes it is retained in the span.

Source reference. Original lemma 06S0: sources/stacks-chow/tags.txt, source lines 8659–8659. Navigation pointer for \ref{spaces-over-fields-lemma-locally-quasi-finite-over-field} used inside the counted span (line 234); the argument that consumes it is retained in the span.

Source reference. Original lemma 0389: sources/stacks-chow/tags.txt, source lines 4228–4228. Navigation pointer for \ref{varieties-lemma-characterize-geometrically-disconnected} used inside the counted span (line 245); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *f is locally of finite type,*
(2) *for every étale morphism $V \to Y$ the map $|X \times_Y V| \to |V|$ is injective.*

Then f is universally injective.

Proof. The question is étale local on Y by Morphisms of Spaces, Lemma 03MX. Hence we may assume that Y is a scheme. Then Y is in particular decent and by Decent Spaces, Lemma 0ACB we see that f is locally quasi-finite. Let $y \in Y$ be a point and let X_y be the scheme theoretic fibre. Assume X_y is not empty. By Spaces over Fields, Lemma 06S0 we see that X_y is a scheme which is locally quasi-finite over $\kappa(y)$. Since $|X_y| \subset |X|$ is the fibre of $|X| \to |Y|$ over y we see that X_y has a unique point x. The same is true for $X_y \times_{\mathrm{Spec}(\kappa(y))} \mathrm{Spec}(k)$ for any finite separable extension $k/\kappa(y)$ because we can realize k as the residue field at a point lying over y in an étale scheme over Y, see More on Morphisms, Lemma 02LF. Thus X_y is geometrically connected, see Varieties, Lemma 0389. This implies that the finite extension $\kappa(x)/\kappa(y)$ is purely inseparable.

We conclude (in the case that Y is a scheme) that for every $y \in Y$ either the fibre X_y is empty, or $(X_y)_{red} = \mathrm{Spec}(\kappa(x))$ with $\kappa(y) \subset \kappa(x)$ purely inseparable. Hence f is radicial (some details omitted), whence universally injective by Lemma 0482. □